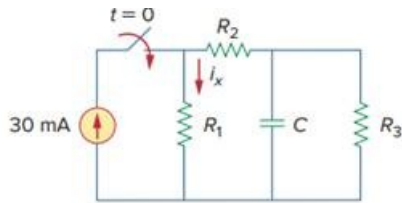


Problem

In the circuit of Fig. 7.117, find i_x for $t > 0$. Let $R_1 = R_2 = 1\text{ k}\Omega$, $R_3 = 2\text{ k}\Omega$, and $C = 0.25\text{ mF}$.

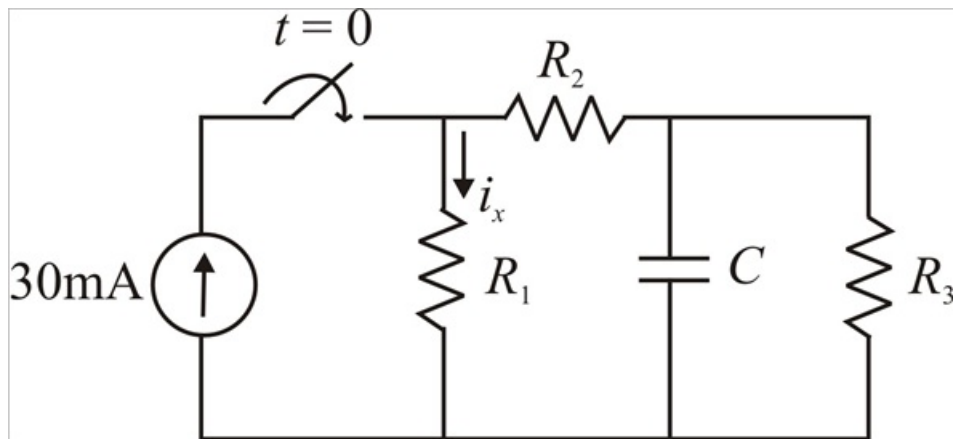
Fig. 7.117:



Step-by-step solution

Step 1 of 11

Given circuit diagram



Step 2 of 11

First we find the voltage across the capacitor

$$v(t) = v(\infty) - [v(0) - v(\infty)]e^{-\frac{t}{\tau}}$$

At $t = 0^-$, there is no source therefore the voltage across the capacitor is

$$v(0^-) = 0\text{V}$$

Since the voltage across a capacitor cannot change instantaneously

$$v(0^+) = v(0^-)$$

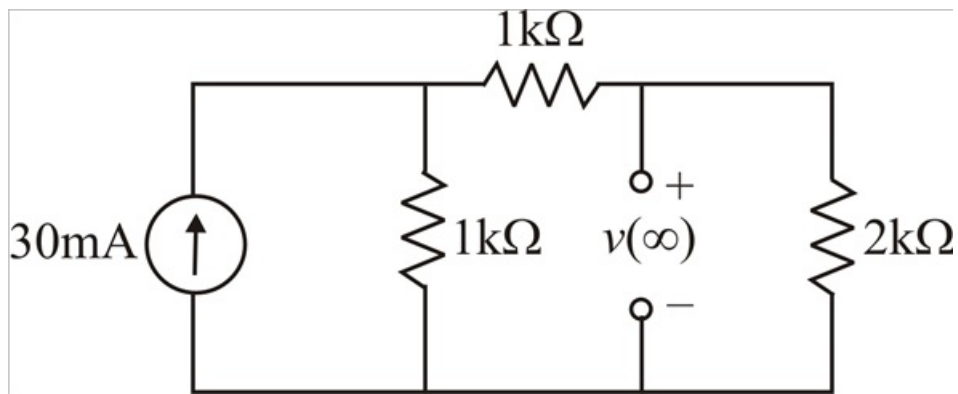
$$v(0^+) = 0\text{V}$$

Therefore the initial voltage across the capacitor is

$$v(0^+) = v(0^-) = v(0) = 0\text{V}$$

Step 3 of 11

At $t = \infty$, the capacitor acts as an open circuit, therefore the voltage across the capacitor is



Step 4 of 11

By using current division rule, current through the $2\text{k}\Omega$ is

$$I_{2\text{k}\Omega} = (30 \times 10^{-3}) \left(\frac{1 \times 10^3}{1 \times 10^3 + 1 \times 10^3 + 2 \times 10^3} \right)$$

$$I_{2\text{k}\Omega} = (30 \times 10^{-3}) \left(\frac{1}{4} \right)$$

$$I_{2\text{k}\Omega} = 7.5 \times 10^{-3}$$

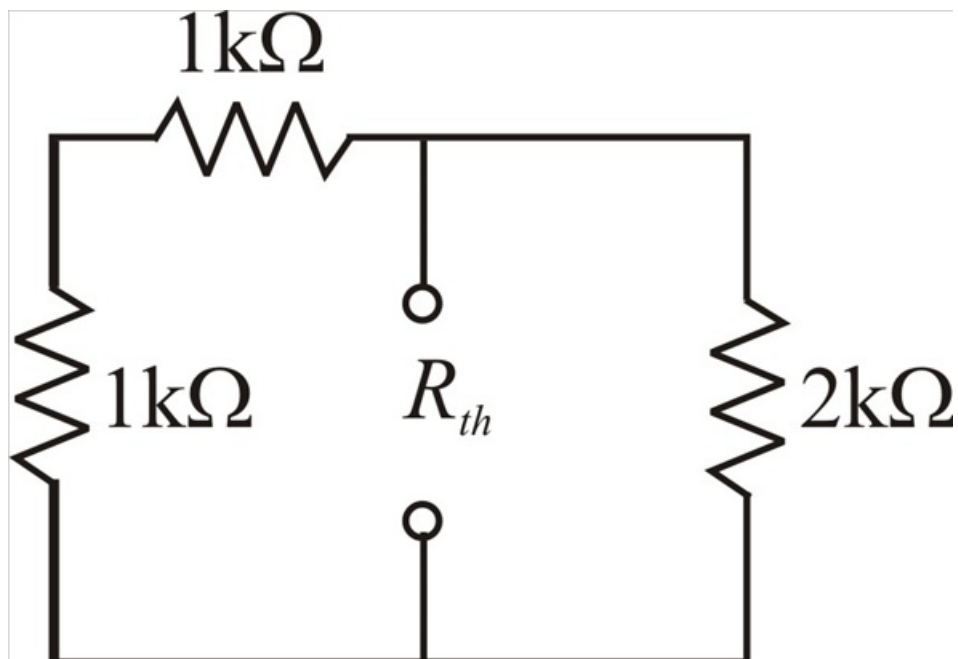
Therefore the voltage across the capacitor is

$$v(\infty) = (7.5 \times 10^{-3}) (2 \times 10^3)$$

$$v(\infty) = 15\text{V}$$

Step 5 of 11

Now the time constant of the circuit is $\tau = R_{th} C$



Step 6 of 11

$$\tau = [(1k + 1k) \parallel 2k] (0.25m)$$

$$\tau = [2k \parallel 2k] (0.25m)$$

$$\tau = \left(\frac{2k \times 2k}{2k + 2k} \right) (0.25m)$$

$$\tau = (1k) (0.25m)$$

$$\tau = (1 \times 10^3) (0.25 \times 10^{-3})$$

$$\tau = 0.25s$$

$$\text{Therefore } \frac{1}{\tau} = 4$$

Step 7 of 11

Therefore the voltage across the capacitor is given by

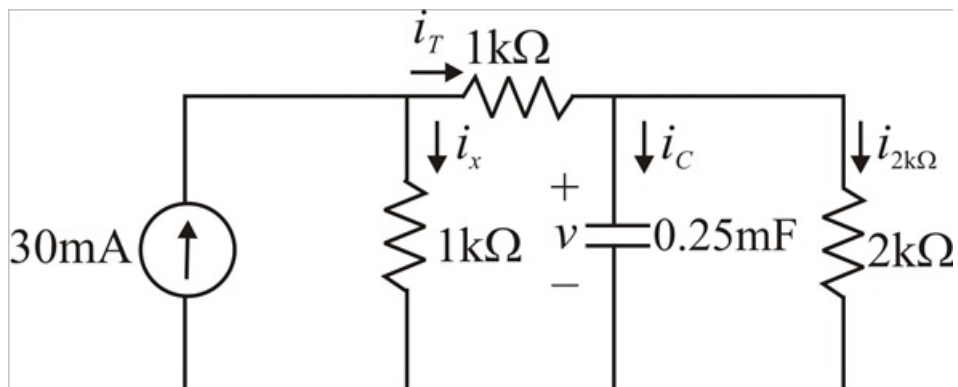
$$v(t) = v(\infty) + [v(0) - v(\infty)] e^{-\frac{t}{\tau}}$$

$$v(t) = 15 + [0 - 15] e^{-\left(\frac{1}{4}\right)t}$$

$$v(t) = (15 - 15e^{-4t}) \text{ V}$$

Step 8 of 11

Now for $t > 0$, the circuit is



Step 9 of 11

Therefore the current through the $2\text{k}\Omega$ is

$$i_{2\text{k}\Omega} = \frac{v(t)}{2 \times 10^3}$$

$$i_{2\text{k}\Omega} = \frac{(15 - 15e^{-4t})}{2 \times 10^3}$$

$$i_{2\text{k}\Omega} = (7.5 - 7.5e^{-4t})\text{mA}$$

The current through the capacitor is

$$i_C = C \frac{dv}{dt}$$

$$i_C = (0.25 \times 10^{-3}) \frac{d}{dt}(15 - 15e^{-4t})$$

$$i_C = (0.25 \times 10^{-3}) [(-15)(-4)e^{-4t}]$$

$$i_C = (0.25 \times 10^{-3}) [60e^{-4t}]$$

$$i_C = 15 \times 10^{-3} e^{-4t}$$

$$i_C = 15e^{-4t}\text{mA}$$

Step 10 of 11

By writing KCL at capacitor node we can write

$$i_T = i_C + i_{2\text{k}\Omega}$$

$$i_T = 15 \times 10^{-3} e^{-4t} + (7.5 - 7.5e^{-4t})10^{-3}$$

$$i_T = 15 \times 10^{-3} e^{-4t} + 7.5 \times 10^{-3} - 7.5 \times 10^{-3} e^{-4t}$$

$$i_T = 7.5 \times 10^{-3} + 7.5 \times 10^{-3} e^{-4t}$$

$$i_T = (7.5 + 7.5e^{-4t})10^{-3}$$

$$i_T = (7.5 + 7.5e^{-4t})\text{mA}$$

Step 11 of 11

By writing KCL at source node we can write

$$30\text{mA} = i_x + i_T$$

$$i_x = 30 \times 10^{-3} - i_T$$

$$i_x = 30 \times 10^{-3} - (7.5 + 7.5e^{-4t})10^{-3}$$

$$i_x = 30 \times 10^{-3} - 7.5 \times 10^{-3} - 7.5 \times 10^{-3} e^{-4t}$$

$$i_x = 22.5 \times 10^{-3} - 7.5 \times 10^{-3} e^{-4t}$$

$$i_x = (22.5 - 7.5e^{-4t})10^{-3}$$

$$\boxed{i_x = 7.5(3 - e^{-4t})\text{mA}}$$